

Презентация

Лабораторная работа №3

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Информация

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Освоить работу с анализатором трафика Wireshark: разобрать кадры Ethernet и изучить PDU транспортного и прикладного уровней стека TCP/IP на реальном трафике.

1. Получить сведения о сетевых интерфейсах через `ipconfig`.
2. Запустить захват трафика в Wireshark на активном интерфейсе.
3. Проверить связь с шлюзом и разобрать ICMP.
4. Разобрать ARP-запросы и ARP-ответы.
5. Пингануть внешние узлы (`rudn.ru`, `ya.ru`) и проанализировать ICMP.
6. Открыть сайт по HTTP и разобрать TCP-сегменты.
7. Изучить DNS и QUIC — работу прикладных протоколов поверх UDP.
8. Разобрать тройное рукопожатие TCP (SYN, SYN+ACK, ACK).

Активное подключение — беспроводная сеть. Видим IPv4/IPv6, маску и шлюз.

```
Адаптер беспроводной локальной сети Подключение по локальной сети* 9:

    Состояние среды. . . . . : Среда передачи недоступна.
    DNS-суффикс подключения . . . . . :

Адаптер беспроводной локальной сети Подключение по локальной сети* 10:

    Состояние среды. . . . . : Среда передачи недоступна.
    DNS-суффикс подключения . . . . . :

Адаптер беспроводной локальной сети Беспроводная сеть:

    DNS-суффикс подключения . . . . . :
    IPv6-адрес. . . . . : fdd5:f158:1a24:0:4d0b:9f55:5f65:7986
    Временный IPv6-адрес. . . . . : fdd5:f158:1a24:0:f937:8ce:945:21bd
    Локальный IPv6-адрес канала . . . : fe80::e1b3:33f6:75fd:b4ae%23
    IPv4-адрес. . . . . : 192.168.1.76
    Маска подсети . . . . . : 255.255.255.0
    Основной шлюз. . . . . : 192.168.1.1

Адаптер Ethernet Сетевое подключение Bluetooth:

    Состояние среды. . . . . : Среда передачи недоступна.
    DNS-суффикс подключения . . . . . :
PS C:\Users\dora2> |
```

MAC 44-a3-bb-57-9b-03, IPv4 192.168.1.76, маска 255.255.255.0, шлюз 192.168.1.1.

```
Адаптер беспроводной локальной сети Беспроводная сеть:  
DNS-суффикс подключения . . . . . :  
Описание . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz  
Физический адрес. . . . . : 44-A3-BB-57-9B-03  
DHCP включен. . . . . : Да  
Автонастройка включена. . . . . : Да  
IPv6-адрес. . . . . : fdd5:f158:1a24:0:4d0b:9f55:5f65:7986(Основной)  
Временный IPv6-адрес. . . . . : fdd5:f158:1a24:0:f937:8ce:945:21bd(Основной)  
Локальный IPv6-адрес канала . . . . : fe80::e1b3:33f6:75fd:b4ae%23(Основной)  
IPv4-адрес. . . . . : 192.168.1.76(Основной)  
Маска подсети . . . . . : 255.255.255.0  
Аренда получена. . . . . : 6 октября 2026 г. 1:54:45  
Срок аренды истекает. . . . . : 6 октября 2026 г. 8:54:45  
Основной шлюз. . . . . : 192.168.1.1  
DHCP-сервер. . . . . : 192.168.1.1  
IAID DHCPv6 . . . . . : 407151547  
DUID клиента DHCPv6 . . . . . : 00-01-00-01-2F-47-EE-9A-34-5A-60-27-02-6B  
DNS-серверы. . . . . : fe80::52ff:20ff:fedd:e4c6%23  
77.88.8.8  
77.88.8.1  
fe80::52ff:20ff:fedd:e4c6%23  
NetBios через TCP/IP. . . . . : Включен
```

Рис. 2: ipconfig /all

Выбор соединения

В Wireshark выбираем активный беспроводной интерфейс для захвата.

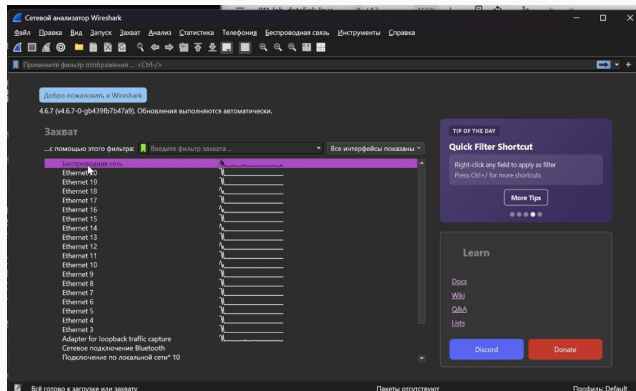


Рис. 3: Выбор соединения

Общий вид окна Wireshark: список пакетов, детали, hex-дамп.

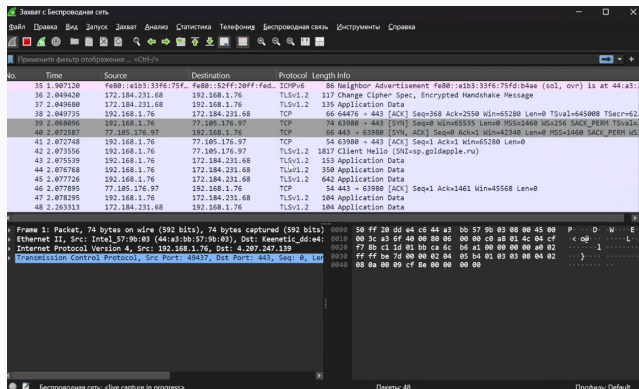


Рис. 4: Wireshark

4 из 4 пакетов, TTL=64, время около 1 мс, потерь нет.

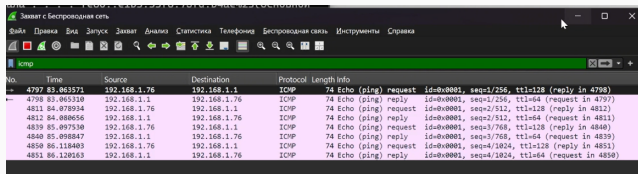
```
PS C:\Users\dora2> ping 192.168.1.1

Обмен пакетами с 192.168.1.1 по 32 байтами данных:
Ответ от 192.168.1.1: число байт=32 время=1мс TTL=64
Ответ от 192.168.1.1: число байт=32 время=1мс TTL=64
Ответ от 192.168.1.1: число байт=32 время=1мс TTL=64
Ответ от 192.168.1.1: число байт=32 время=1мс TTL=64

Статистика Ping для 192.168.1.1:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)
Приблизительное время приема-передачи в мс:
    Минимальное = 1мсек, Максимальное = 1 мсек, Среднее = 1 мсек
PS C:\Users\dora2>
```

Рис. 5: ping

8 строк: 4 отправки и 4 ответа между 192.168.1.76 и 192.168.1.1.



The screenshot shows a Wireshark capture of ICMP traffic. The packet list pane displays 8 packets, alternating between requests and replies. The packet details pane shows the structure of an ICMP Echo (ping) packet, including the ID, sequence number, and TTL.

No.	Time	Source	Destination	Protocol	Length	Info
→ 4797	83.063571	192.168.1.1	192.168.1.76	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 4798)
← 4798	83.065310	192.168.1.1	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 4797)
→ 4811	84.078934	192.168.1.76	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 4812)
← 4812	84.080656	192.168.1.1	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 4811)
→ 4839	85.097530	192.168.1.76	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 4840)
← 4840	85.098847	192.168.1.1	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 4839)
→ 4850	86.118403	192.168.1.76	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 4851)
← 4851	86.120163	192.168.1.1	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=64 (request in 4850)

Рис. 6: icmp

Кадр 74 байта, Ethernet II, Intel → Keentic, внутри IPv4 и ICMP.

```

▼ Frame 238: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{9BEDC1B5-0CE5-486D-9D72-87D28CF4E4C8}, id 0
  Section number: 1
  > Interface id: 0 (\Device\NPF_{9BEDC1B5-0CE5-486D-9D72-87D28CF4E4C8})
    Encapsulation type: Ethernet (1)
    Arrival Time: Oct  6, 2026 02:25:01.926642000 RTZ 2 (amma)
    UTC Arrival Time: Oct  5, 2026 23:25:01.926642000 UTC
    Epoch Arrival Time: 1791242701.926642000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frames: 81.432000 milliseconds]
    [Time since reference or first frame: 33.799034000 seconds]
    Frame Number: 238
    Frame Length: 74 bytes (592 bits)
    Capture Length: 74 bytes (592 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:icmp:data]
    Character encoding: ASCII (0)
    [Coloring Rule Name: ICMP]
    [Coloring Rule String: icmp || icmpv6]
  > Ethernet II, Src: Intel_57:9b:03 (44:a3:bb:57:9b:03), Dst: Keenetic_dd:e4:c6 (58:ff:20:dd:e4:c6)
    > Destination: Keenetic_dd:e4:c6 (58:ff:20:dd:e4:c6)
      ....0. .... = LG bit: Globally unique address (factory default)
      ....0. .... = IG bit: Individual address (unicast)
    > Source: Intel_57:9b:03 (44:a3:bb:57:9b:03)
      Type: IPv4 (8x8000)
      [Stream index: 1]
    > Internet Protocol Version 4, Src: 192.168.1.76, Dst: 192.168.1.1
    > Internet Control Message Protocol

```

Рис. 7: Исходящий ICMP

MAC-адреса стоят в обратном порядке: 50:ff:20:dd:e4:c6 → 44:a3:bb:57:9b:03.

```
> Frame 231: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{9BEDC1B5-0CE5-406D-9072-87820CF4E4C8}, Id 0
  Section number: 1
  > Interface id: 0 (\Device\NPF_{9BEDC1B5-0CE5-406D-9072-87820CF4E4C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Oct 6, 2025 02:25:01.928430000 RTZ 2 (sma)
  UTC Arrival Time: Oct 5, 2025 23:25:01.928430000 UTC
  Epoch Arrival Time: 1791242701.928430000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 1.788000 milliseconds]
  [Time delta from previous displayed frame: 1.788000 milliseconds]
  [Time since reference or first frame: 33.800022000 seconds]
  Frame Number: 231
  Frame Length: 74 bytes (592 bits)
  Capture Length: 74 bytes (592 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:icmp:data]
  Character encoding: ASCII (0)
  [Coloring Rule Name: ICMP]
  [Coloring Rule String: icmp || icmpv6]
  > Ethernet II, Src: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6), Dst: Intel_57:9b:03 (44:a3:bb:57:9b:03)
    > Destination: Intel_57:9b:03 (44:a3:bb:57:9b:03)
      ....0. .... = 16 bit: Globally unique address (factory default)
      ....0. .... = 16 bit: Individual address (unicast)
    > Source: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
      ....0. .... = 16 bit: Globally unique address (factory default)
      ....0. .... = 16 bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 1]
  > Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.76
  > Internet Control Message Protocol
```

Рис. 8: Входящий ICMP

Who has 192.168.1.76? Tell 192.168.1.1 — шлюз ищет наш MAC.

No.	Time	Source	Destination	Protocol	Length	Info
229	33.717682	Intel_57:9b:03	Keenetic_dd:e4:c6	ARP	42	192.168.1.76 is at 44:a3:bb:57:9b:03
52	5.942725	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.149? Tell 192.168.1.1
65	7.884865	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.43? Tell 192.168.1.1
164	27.340617	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
166	28.373390	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
167	28.373390	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
168	28.373390	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
185	29.280850	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
186	29.309308	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
196	30.312166	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.75? Tell 192.168.1.1
197	30.413803	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
204	31.335563	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
228	33.717586	Keenetic_dd:e4:c6	Intel_57:9b:03	ARP	42	Who has 192.168.1.76? Tell 192.168.1.1
Frame 228: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{98EDC1B5-0CE5-4060-9072-87020CF4E4C8}, id 0 Section number: 1 Interface id: 0 (\Device\NPF_{98EDC1B5-0CE5-4060-9072-87020CF4E4C8}) Encapsulation type: Ethernet (1) Arrival Time: Oct 6, 2026 02:25:01.845194000 RTZ 2 (zma) UTC Arrival Time: Oct 6, 2026 23:25:01.845194000 UTC Epoch Arrival Time: 1791242701.845194000 [Time shift for this packet: 0.000000000 seconds] [Time delta from previous captured frame: 92.4099000 milliseconds] [Time delta from previous displayed frame: 2.382038000 seconds] [Time since reference or first frame: 33.717586000 seconds] Frame Number: 228 Frame Length: 42 bytes (336 bits) Capture Length: 42 bytes (336 bits) [Frame is marked: False] [Frame is ignored: False] [Protocols in frame: ethertype:arp] Character encoding: ASCII (0) [Coloring Rule Name: ARP] [Coloring Rule String: arp] Ethernet II, Src: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6), Dst: Intel_57:9b:03 (44:a3:bb:57:9b:03) Destination: Intel_57:9b:03 (44:a3:bb:57:9b:03) 0 = LG bit: Globally unique address (factory default) 0 = IG bit: Individual address (unicast) Source: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6) 0 = LG bit: Globally unique address (factory default) 0 = IG bit: Individual address (unicast) Type: ARP (0x0006) [Stream index: 1] Address Resolution Protocol (request)						

Рис. 9: ARP-запрос

192.168.1.76 is at 44:a3:bb:57:9b:03 — тип reply.

```

229 31.717602 Intel_57:9b:03 Keenetic_dd:e4:c6 ARP 42 192.168.1.76 is at 44:a3:bb:57:9b:03
52 5.942725 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.149? Tell 192.168.1.1
65 7.884865 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.43? Tell 192.168.1.1
164 27.340617 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.73? Tell 192.168.1.1
166 28.373390 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.69? Tell 192.168.1.1
167 28.373390 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.73? Tell 192.168.1.1
168 28.373390 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.67? Tell 192.168.1.1
185 29.286850 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.73? Tell 192.168.1.1
186 29.389388 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.67? Tell 192.168.1.1
196 30.312166 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.73? Tell 192.168.1.1
197 30.413803 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.67? Tell 192.168.1.1
204 31.335563 Keenetic_dd:e4:c6 Broadcast ARP 42 who has 192.168.1.67? Tell 192.168.1.1
228 33.717586 Keenetic_dd:e4:c6 Intel_57:9b:03 ARP 42 who has 192.168.1.76? Tell 192.168.1.1

+ Frame 229: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{9BEDC1B5-8CE5-406D-9072-B7020CF4E4C8}, id 0
  Section number: 1
  + Interface id: 0 (\Device\NPF_{9BEDC1B5-8CE5-406D-9072-B7020CF4E4C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Oct 6, 2026 02:25:01.845210000 RTZ 2 (sma)
  UTC Arrival Time: Oct 5, 2026 23:25:01.845210000 UTC
  Epoch Arrival Time: 1791242781.845210000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 16.000 microseconds]
  [Time delta from previous displayed frame: 16.000 microseconds]
  [Time since reference or first frame: 33.717602000 seconds]
  Frame Number: 229
  Frame Length: 42 bytes (336 bits)
  Capture Length: 42 bytes (336 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:arp]
  Character encoding: ASCII (0)
  [Coloring Rule Name: ARP]
  [Coloring Rule String: arp]
+ Ethernet II, Src: Intel_57:9b:03 (44:a3:bb:57:9b:03), Dst: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
+ Destination: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
  ....0. .... = LG bit: Globally unique address (factory default)
  ....0. .... = IG bit: Individual address (unicast)
+ Source: Intel_57:9b:03 (44:a3:bb:57:9b:03)
  ....0. .... = LG bit: Globally unique address (factory default)
  ....0. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 1]
+ Address Resolution Protocol (reply)

```

Рис. 10: ARP-ответ

rudn.ru — 100 % потерь, ya.ru (77.88.44.242) — 4 из 4.

```
PS C:\Users\dora2> ping rudn.ru

Обмен пакетами с rudn.ru [37.18.93.135] с 32 байтами данных:
Превышен интервал ожидания для запроса.
Превышен интервал ожидания для запроса.
Превышен интервал ожидания для запроса.
Превышен интервал ожидания для запроса.

Статистика Ping для 37.18.93.135:
    Пакетов: отправлено = 4, получено = 0, потеряно = 4
    (100% потерь)
PS C:\Users\dora2> ping ya.ru

Обмен пакетами с ya.ru [77.88.44.242] с 32 байтами данных:
Ответ от 77.88.44.242: число байт=32 время=5мс TTL=50
Ответ от 77.88.44.242: число байт=32 время=5мс TTL=50
Ответ от 77.88.44.242: число байт=32 время=5мс TTL=50
Ответ от 77.88.44.242: число байт=32 время=5мс TTL=50

Статистика Ping для 77.88.44.242:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)
Приблизительное время приема-передачи в мс:
    Минимальное = 5мсек, Максимальное = 5 мсек, Среднее = 5 мсек
PS C:\Users\dora2>
```

Рис. 11: ping rudn.ru и ya.ru

Снова запрос Who has 192.168.1.76? и ответ 192.168.1.76 is at 44:a3:bb:57:9b:03.

No	Time	Source	Destination	Protocol	Length	Info
327	17.531974	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.76? Tell 192.168.1.1
328	17.531912	Keenetic_dd:e4:c6	Intel_57:9b:03	ARP	42	Who has 192.168.1.76? Tell 192.168.1.1
327	17.531971	Intel_57:9b:03	Keenetic_dd:e4:c6	ARP	42	192.168.1.76 is at 44:a3:bb:57:9b:03
484	38.397837	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
496	39.311313	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
502	40.343381	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
503	40.437779	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
514	41.256987	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
523	44.413148	Keenetic_dd:e4:c6	Intel_57:9b:03	ARP	42	Who has 192.168.1.76? Tell 192.168.1.1
524	44.413198	Intel_57:9b:03	Keenetic_dd:e4:c6	ARP	42	192.168.1.76 is at 44:a3:bb:57:9b:03


```

- Frame 326: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{9BEDC185-0CE5-406D-9D72-87828CFAE4C8}, id 0
  Section number: 1
  > Interface id: 0 (\Device\NPF_{9BEDC185-0CE5-406D-9D72-87828CFAE4C8})
    Incapsulation type: Ethernet (1)
    Arrival Time: Oct 6, 2026 02:30:17.237678000 RTZ 2 (swua)
    UTC Arrival Time: Oct 5, 2026 23:30:17.237678000 UTC
    Epoch Arrival Time: 175123017.237678000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 170.654000 milliseconds]
    [Time delta from previous displayed frame: 12.320956000 seconds]
    [Time since reference or first frame: 17.531920000 seconds]
    Frame Number: 326
    Frame Length: 42 bytes (336 bits)
    Capture Length: 42 bytes (336 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:arp]
    Character encoding: ASCII (0)
    [Coloring Rule Name: ARP]
    [Coloring Rule String: arp]
  > Ethernet II, Src: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6), Dst: Intel_57:9b:03 (44:a3:bb:57:9b:03)
    > Destination: Intel_57:9b:03 (44:a3:bb:57:9b:03)
      ....0. .... = LG bit: Globally unique address (factory default)
      ....0. .... = IG bit: Individual address (unicast)
    > Source: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
      ....0. .... = LG bit: Globally unique address (factory default)
      ....0. .... = IG bit: Individual address (unicast)
    Type: ARP (0x8006)
    [Stream index: 0]
  > Address Resolution Protocol (request)
  
```

Рис. 12: ARP для ya.ru

Отличается только направлением, тип reply.

No.	Time	Source	Destination	Protocol	Length	Info
18	5.210976	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.67? Tell 192.168.1.1
326	17.531932	Keenetic_dd:e4:c6	Intel_57:9b:03	ARP	42	Who has 192.168.1.76? Tell 192.168.1.1
327	17.531971	Intel_57:9b:03	Keenetic_dd:e4:c6	ARP	42	192.168.1.76 is at 44:a3:bb:57:9b:03
484	38.397837	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
486	39.311313	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
502	40.343381	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
503	40.437779	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
514	41.256987	Keenetic_dd:e4:c6	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.1
523	44.413148	Keenetic_dd:e4:c6	Intel_57:9b:03	ARP	42	Who has 192.168.1.76? Tell 192.168.1.1
524	44.413198	Intel_57:9b:03	Keenetic_dd:e4:c6	ARP	42	192.168.1.76 is at 44:a3:bb:57:9b:03


```
> Frame 327: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{9BEDC1B5-0CE5-406D-9072-87028CF4E4C8}, id 0
  Section number: 1
  > Interface id: 0 (\Device\NPF_{9BEDC1B5-0CE5-406D-9072-87028CF4E4C8})
    Encapsulation type: Ethernet (1)
    Arrival Time: Oct 6, 2026 02:30:17.237717000 RTZ 2 (smva)
    UTC Arrival Time: Oct 5, 2026 23:30:17.237717000 UTC
    Epoch Arrival Time: 1751243017.237717000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 39.000 microseconds]
    [Time delta from previous displayed frame: 39.000 microseconds]
    [Time since reference or first frame: 17.531971000 seconds]
    Frame Number: 327
    Frame Length: 42 bytes (336 bits)
    Capture Length: 42 bytes (336 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:arp]
    Character encoding: ASCII (0)
    [Coloring Rule Name: ARP]
    [Coloring Rule String: arp]
  > Ethernet II, Src: Intel_57:9b:03 (44:a3:bb:57:9b:03), Dst: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
    = Destination: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
      ....0. .... = LG bit: Globally unique address (factory default)
      ....0. .... = IG bit: Individual address (unicast)
    = Source: Intel_57:9b:03 (44:a3:bb:57:9b:03)
      ....0. .... = LG bit: Globally unique address (factory default)
      ....0. .... = IG bit: Individual address (unicast)
    Type: ARP (0x0806)
    [Stream index: 0]
  > Address Resolution Protocol (reply)
```

Рис. 13: Второй ARP

На сетевом уровне адрес 77.88.44.242, на Ethernet — по-прежнему шлюз.

No.	Time	Source	Destination	Protocol	Length	Info
346	23.983268	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=21/5376, ttl=128 (no response found)
371	28.614618	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=22/5632, ttl=128 (no response found)
444	31.616253	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=23/5888, ttl=128 (no response found)
491	38.619478	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=24/6144, ttl=128 (no response found)
561	50.059113	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=25/6400, ttl=128 (reply in 562)
562	50.055377	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=25/6400, ttl=50 (request in 561)
567	51.002456	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=26/6656, ttl=128 (reply in 568)
568	51.067700	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=26/6656, ttl=50 (request in 567)
570	52.008488	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 572)
572	52.005955	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=27/6912, ttl=50 (request in 570)
575	53.005587	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 576)
576	53.001000	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=28/7168, ttl=50 (request in 575)


```

+ Frame 561: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{98EDC185-0CE5-486D-9072-87020CF4E4C8}, id 0
  Section number: 1
    + Interface id: 0 (\Device\NPF_{98EDC185-0CE5-486D-9072-87020CF4E4C8})
    Encapsulation type: ethernet (1)
    Arrival Time: Oct  6, 2026 02:30:49.755859000 RTZ 2 (swm)
    UTC Arrival Time: Oct  5, 2026 23:30:49.755859000 UTC
    Epoch Arrival Time: 1791243049.755859000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 3.152000 milliseconds]
    [Time delta from previous displayed frame: 11.430633000 seconds]
    [Time since reference or first frame: 50.050113000 seconds]
    Frame Number: 561
    Frame Length: 74 bytes (592 bits)
    Capture Length: 74 bytes (592 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:icmp:data]
    Character encoding: ASCII (0)
    [Coloring Rule Name: ICMP]
    [Coloring Rule String: icmp || icmpv6]
  + Ethernet II, Src: Intel_57:9b:03 (44:a3:bb:57:9b:03), Dst: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
    + Destination: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
      ....0. .... = 10 bit: Globally unique address (factory default)
      ....0. .... = 16 bit: Individual address (unicast)
    + Source: Intel_57:9b:03 (44:a3:bb:57:9b:03)
      ....0. .... = 10 bit: Globally unique address (factory default)
      ....0. .... = 16 bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 0]
  + Internet Protocol Version 4, Src: 192.168.1.76, Dst: 77.88.44.242
  + Internet Control Message Protocol
  
```

Рис. 14: ICMP к ya.ru

Источник 77.88.44.242, получатель 192.168.1.76.

No.	Time	Source	Destination	Protocol	Length	Info
346	23.983268	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=21/5376, ttl=128 (no response found)
371	28.614618	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=22/5631, ttl=128 (no response found)
444	33.616253	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=23/5888, ttl=128 (no response found)
491	38.619478	192.168.1.76	37.18.93.135	ICMP	74	Echo (ping) request id=0x0001, seq=24/6144, ttl=128 (no response found)
561	50.055113	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=25/6400, ttl=128 (reply in 562)
562	50.055172	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=25/6400, ttl=58 (request in 561)
567	51.062456	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=26/6656, ttl=128 (reply in 568)
568	51.067700	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=26/6656, ttl=58 (request in 567)
570	52.008488	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 572)
572	52.008595	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=27/6912, ttl=58 (request in 570)
575	53.005587	192.168.1.76	77.88.44.242	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 576)
576	53.009180	77.88.44.242	192.168.1.76	ICMP	74	Echo (ping) reply id=0x0001, seq=28/7168, ttl=58 (request in 575)


```

+ Frame 562: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{98EDC1B5-0CE5-4060-9072-87020CF4E4C8}, id 0
  Section number: 1
  + Interface id: 0 (\Device\NPF_{98EDC1B5-0CE5-4060-9072-87020CF4E4C8})
  + Encapsulation type: Ethernet (1)
  + Arrival Time: Oct  5, 2025 02:30:49.761123000 RTZ 2 (n/a)
  + UTC Arrival Time: Oct  5, 2025 23:30:49.761123000 UTC
  + Epoch Arrival Time: 1791243849.761123000
  + [Time shift for this packet: 0.000000000 seconds]
  + [Time delta from previous captured frame: 5.264000 milliseconds]
  + [Time delta from previous displayed frame: 5.264000 milliseconds]
  + [Time since reference or first frame: 50.055177000 seconds]
  + Frame Number: 562
  + Frame Length: 74 bytes (592 bits)
  + Capture Length: 74 bytes (592 bits)
  + [Frame is marked: False]
  + [Frame is ignored: False]
  + [Protocols in frame: ethernet II: icmp: data]
  + Character encoding: ASCII (0)
  + [Coloring Rule Name: ICMP]
  + [Coloring Rule String: icmp || icmpv6]
  + Ethernet II, Src: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6), Dst: Intel_57:9b:03 (44:a3:b0:57:9b:03)
    + Destination: Intel_57:9b:03 (44:a3:b0:57:9b:03)
      ....0. .... = LG bit: Globally unique address (factory default)
      ....0. .... = IG bit: Individual address (unicast)
    + Source: Keenetic_dd:e4:c6 (50:ff:20:dd:e4:c6)
      ....0. .... = LG bit: Globally unique address (factory default)
      ....0. .... = IG bit: Individual address (unicast)
    + Type: IPv6 (0x0000)
    + [Stream index: 0]
  + Internet Protocol Version 4, Src: 77.88.44.242, Dst: 192.168.1.76
  + Internet Control Message Protocol
    
```

Рис. 15: Ответ от ya.ru

Открываем info.cern.ch по http поверх TCP.

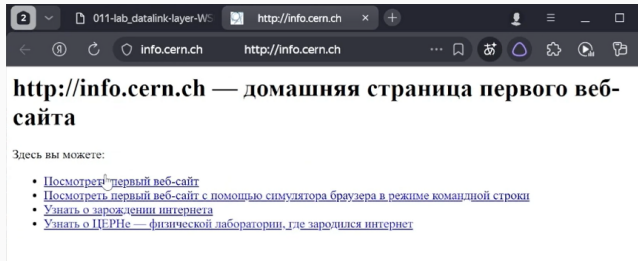


Рис. 16: HTTP-подключение

GET /... HTTP/1.1, порт 52037 → 80, TCP с флагами PSH, ACK.

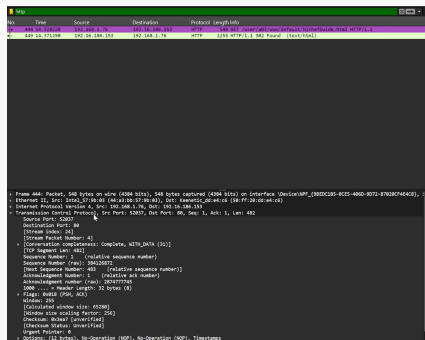
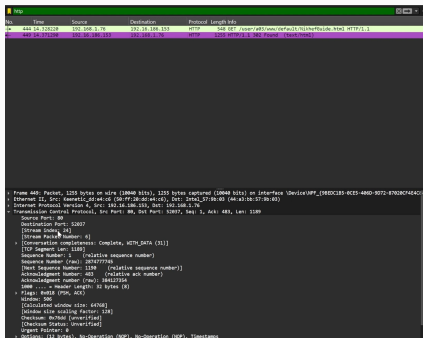


Рис. 17: Исходящий HTTP

Источник 192.16.186.153:80, ответ HTTP/1.1 302 Found, длина 1189.



```

No.    Time           Source                Destination            Protocol Length Info
14    444.14-124.128  192.16.186.153       192.16.186.154        HTTP      548 GET /user/493/www.cerfa.li/likefacebook.html HTTP/1.1
     444.14-124.128  192.16.186.154       192.16.186.154        HTTP      1189 HTTP/1.1 302 Found - [redirected]

...

Frame 445: Packet, 1235 bytes on wire (10040 bits), 1235 bytes captured (10040 bits) on interface DeviceUPP_198DC185-6CE5-488D-9072-87028CF4640A
Ethernet II, Src: Realtek-8d4e4c (58:ff:38:d6:4e:4c), Dst: Intel-57:9d:63 (44:ad:b5:57:9d:63)
Internet Protocol Version 4, Src: 192.16.186.153, Dst: 192.16.186.154
Transmission Control Protocol, Src Port: 80, Dst Port: 52807, Seq: 1, Ack: 483, Len: 1189
  Source Port: 80
  Destination Port: 52807
  [Stream index: 24]
  [Stream PathID Number: 6]
  [Computation completeness: Complete, HTTP_DATA (31)]
  [TCP Segment Len: 1189]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 261277798
  [Next Sequence Number: 1190 (relative sequence number)]
  Acknowledgment Number: 483 (relative ack number)
  Acknowledgment number (raw): 386127314
  1189 ... => Header length: 32 bytes (8)
  Flags: RST, FIN, ACK
  Window: 586
  [Calculated window size: 64768]
  Window size scaling factor: 328
  Checksum: 0x76dd [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (11 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
```

Рис. 18: Входящий HTTP

DNS поверх UDP, порт источника 53, сервер 77.88.8.1.

No.	Time	Source	Destination	Protocol	Length	Info
408	13.883109	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xbc59 HTTPS nic.nikhef.nl
409	13.883591	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x10a4 A nic.nikhef.nl
410	13.995083	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xf083 HTTPS nic.nikhef.nl
411	13.995371	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x453b A nic.nikhef.nl
412	14.000391	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xaa91 HTTPS nic.nikhef.nl
413	14.000584	192.168.1.76	77.88.8.1	DNS	73	Standard query 0x9e17 A nic.nikhef.nl
419	14.092002	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x10a4 A nic.nikhef.nl CNAME www.nikhef.nl C
420	14.119093	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0xbc59 HTTPS nic.nikhef.nl CNAME www.nikhef.
424	14.189724	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x453b A nic.nikhef.nl CNAME www.nikhef.nl C
425	14.201846	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x9e17 A nic.nikhef.nl CNAME www.nikhef.nl C
426	14.210008	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0xf083 HTTPS nic.nikhef.nl CNAME www.nikhef.
429	14.223595	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0xaa91 HTTPS nic.nikhef.nl CNAME www.nikhef.
450	14.373630	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xdf80 HTTPS www.nikhef.nl
451	14.373765	192.168.1.76	77.88.8.1	DNS	73	Standard query 0xa274 A www.nikhef.nl
453	14.506532	77.88.8.1	192.168.1.76	DNS	113	Standard query response 0xa274 A www.nikhef.nl CNAME lvp029web.nikhe
457	14.576970	77.88.8.1	192.168.1.76	DNS	163	Standard query response 0xdf80 HTTPS www.nikhef.nl CNAME lvp029web.n
519	15.044930	192.168.1.76	77.88.8.1	DNS	89	Standard query 0x0501 A ya.ru
527	15.045488	192.168.1.76	77.88.8.1	DNS	112	Standard query 0xed6c A browser.translate.yandex.net
529	15.045606	192.168.1.76	77.88.8.1	DNS	112	Standard query 0x5c55 HTTPS browser.translate.yandex.net
531	15.045715	192.168.1.76	77.88.8.1	DNS	89	Standard query 0xc928 HTTPS ya.ru
534	15.050697	77.88.8.1	192.168.1.76	DNS	139	Standard query response 0x0501 A ya.ru A 5.255.255.242 A 77.88.44.24
539	15.051745	77.88.8.1	192.168.1.76	DNS	139	Standard query response 0xed6c A browser.translate.yandex.net A 87.2
543	15.051745	77.88.8.1	192.168.1.76	DNS	165	Standard query response 0x5c55 HTTPS browser.translate.yandex.net 50
544	15.051745	77.88.8.1	192.168.1.76	DNS	147	Standard query response 0xc928 HTTPS ya.ru SOA ns1.yandex.ru
622	15.235544	192.168.1.76	77.88.8.1	DNS	105	Standard query 0x2008 HTTPS api.browser.yandex.ru

Frame 457: Packet, 163 bytes on wire (1304 bits), 163 bytes captured (1304 bits) on interface \Device\NPF_{90EDC1B5-0CE5-4060-9D72-87020CF4E4C8}, id 44:a3:bb:57:9b:03

Ethernet II, Src: Keenetic_dd:e4:c6 (58:ff:20:dd:e4:c6), Dst: Intel_57:9b:03 (44:a3:bb:57:9b:03)

Internet Protocol Version 4, Src: 77.88.8.1, Dst: 192.168.1.76

User Datagram Protocol, Src Port: 53, Dst Port: 49664

Source Port: 53

Destination Port: 49664

Length: 129

Checksum: 0x5e00 [Unverified]

[Checksum Status: Unverified]

[Stream index: 8]

[Stream Packet Number: 2]

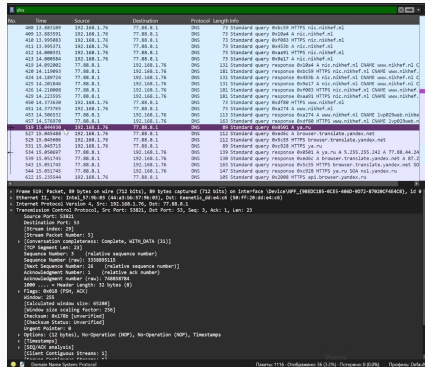
[Timestamp]

UDP payload (121 bytes)

Domain Name System (response)

Рис. 19: Исходящий DNS

Порты 53 → 49664, отличаются длина и контрольная сумма.



No.	Time	Source	Destination	Protocol	Length	Info
408	13.801029	192.168.1.76	77.88.8.1	DNS	72	Standard query 0x0010 HTTPS rsc.nishref.nl
409	13.803591	192.168.1.76	77.88.8.1	DNS	70	Standard query 0x0044 A rsc.nishref.nl
410	13.905883	192.168.1.76	77.88.8.1	DNS	70	Standard query 0x0003 HTTPS rsc.nishref.nl
411	13.965571	192.168.1.76	77.88.8.1	DNS	70	Standard query 0x0010 A rsc.nishref.nl
412	14.000031	192.168.1.76	77.88.8.1	DNS	70	Standard query 0x0001 HTTPS rsc.nishref.nl
413	14.000554	192.168.1.76	77.88.8.1	DNS	70	Standard query 0x0017 A rsc.nishref.nl
419	14.002082	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x0044 A rsc.nishref.nl CNAME www.nishref.ru C
420	14.110933	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0x0039 HTTPS rsc.nishref.nl CNAME www.nishref
424	14.189724	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x0010 A rsc.nishref.nl CNAME www.nishref.ru C
425	14.209340	77.88.8.1	192.168.1.76	DNS	131	Standard query response 0x0017 A rsc.nishref.nl CNAME www.nishref.ru C
426	14.210000	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0x0003 HTTPS rsc.nishref.nl CNAME www.nishref
429	14.223595	77.88.8.1	192.168.1.76	DNS	181	Standard query response 0x0001 HTTPS rsc.nishref.nl CNAME www.nishref
436	14.337640	192.168.1.76	77.88.8.1	DNS	70	Standard query 0x0000 HTTPS www.nishref.nl
433	14.377563	192.168.1.76	77.88.8.1	DNS	70	Standard query 0x0074 A www.nishref.nl
435	14.500532	77.88.8.1	192.168.1.76	DNS	113	Standard query response 0x0074 A www.nishref.nl CNAME lyp8cheb.nishk
437	14.570709	77.88.8.1	192.168.1.76	DNS	103	Standard query response 0x0000 HTTPS www.nishref.nl CNAME lyp8cheb-n
519	15.044130	192.168.1.76	77.88.8.1	DNS	80	Standard query 0x0001 A ya.ru
517	15.045430	192.168.1.76	77.88.8.1	DNS	113	Standard query response 0x0001 A browser.translate.yandex.net
529	15.045880	192.168.1.76	77.88.8.1	DNS	102	Standard query 0x0035 HTTPS browser.translate.yandex.net
531	15.045712	192.168.1.76	77.88.8.1	DNS	80	Standard query 0x0020 HTTPS ya.ru
534	15.050697	77.88.8.1	192.168.1.76	DNS	100	Standard query response 0x0001 A ya.ru A 5.255.255.242 A 77.88.44.34
539	15.051745	77.88.8.1	192.168.1.76	DNS	100	Standard query response 0x0010 A browser.translate.yandex.net A 87.2
543	15.051745	77.88.8.1	192.168.1.76	DNS	100	Standard query response 0x0035 HTTPS browser.translate.yandex.net SO
544	15.051745	77.88.8.1	192.168.1.76	DNS	147	Standard query response 0x0020 HTTPS ya.ru SOA ns1.yandex.ru
622	15.235544	192.168.1.76	77.88.8.1	DNS	100	Standard query 0x0000 HTTPS api.browsers.yandex.ru

Frame 529: Packet, 80 bytes on wire (712 bits), 80 bytes captured (712 bits) on Interface [Device\NPF_{...}]

Ethernet II, Src: Intel(R) Gigabit Ethernet Controller (82:5B:64:00:00:00), Dst: Intel(R) Gigabit Ethernet Controller (82:5B:64:00:00:00)

Internet Protocol Version 4, Src: 192.168.1.76, Dst: 77.88.8.1

Transmission Control Protocol, Src Port: 58821, Dst Port: 53, Seq: 3, Ack: 1, Len: 23

Source Port: 58821

Destination Port: 53

[Stream Index: 20]

[Stream Packet Number: 5]

[Conversation Summary: Complete, NTH_DATA (32)]

[TCP Segment Len: 23]

Sequence Number: 3 (relative sequence number)

Sequence Number (raw): 310809113

[Next Sequence Number: 38 (relative sequence number)]

Acknowledgment Number: 3 (relative seq. number)

Acknowledgment Number (raw): 748838764

Jump ... < Header Length: 32 bytes (8)

Flags: Reset (RST, ACK)

Window: 255

[Calculated window size: 65536]

[Window size scaling factor: 256]

Checksum: 0x17B [Unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

Options: (22 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps

[Timestamps]

[TCP-NV analysis]

[Client Contiguous Streams: 1]

[Security associations: none detected]

Display Name System Protocol

Пакеты: 1118 - Отображение: 16 (1.4%) - Фильтры: 0 (0.0%) - Показать: Default

Рис. 20: Входящий DNS

Исходящий QUIC

QUIC поверх UDP: 54581 → 443.

No.	Time	Source	Destination	Protocol	Length	Info
80	0.000	192.168.1.70	172.217.110.4	QUIC	1200	Initial, DCID=407b1e53b0b9e5c5, PN: 1, CRYPTO
81	0.000	192.168.1.70	172.217.110.4	QUIC	1200	Initial, DCID=407b1e53b0b9e5c5, PN: 2, CRYPTO
82	0.000	192.168.1.70	172.217.110.4	QUIC	1200	Initial, DCID=407b1e53b0b9e5c5, PN: 3, CRYPTO, CRYPTO, PROTECTED_PAYLOAD, CRYPTO...
83	0.000	192.168.1.70	172.217.110.4	QUIC	121	ACK, DCID=407b1e53b0b9e5c5, PN: 1, ACK
84	0.000	192.168.1.70	172.217.110.4	QUIC	82	Initial, DCID=407b1e53b0b9e5c5, PN: 2, ACK
85	0.000	192.168.1.70	172.217.110.4	QUIC	82	Initial, DCID=407b1e53b0b9e5c5, PN: 3, ACK, PROTECTED_PAYLOAD
86	0.000	192.168.1.70	172.217.110.4	QUIC	1200	Initial, DCID=407b1e53b0b9e5c5, PN: 4, ACK, PROTECTED_PAYLOAD
87	0.000	192.168.1.70	172.217.110.4	QUIC	1200	Protected Payload (PP)
88	0.000	192.168.1.70	172.217.110.4	QUIC	80	Protected Payload (PP)
89	0.000	192.168.1.70	172.217.110.4	QUIC	208	Protected Payload (PP)
90	0.000	192.168.1.70	172.217.110.4	QUIC	136	Protected Payload (PP)
91	0.000	192.168.1.70	172.217.110.4	QUIC	73	Protected Payload (PP)
92	0.000	192.168.1.70	172.217.110.4	QUIC	1208	Protected Payload (PP)
93	0.000	192.168.1.70	172.217.110.4	QUIC	108	Protected Payload (PP)
94	0.000	192.168.1.70	172.217.110.4	QUIC	88	Protected Payload (PP)
95	0.000	192.168.1.70	172.217.110.4	QUIC	102	Protected Payload (PP)
96	0.000	192.168.1.70	172.217.110.4	QUIC	73	Protected Payload (PP)
97	0.000	192.168.1.70	172.217.110.4	QUIC	67	Protected Payload (PP)
98	0.000	192.168.1.70	172.217.110.4	QUIC	78	Protected Payload (PP)
99	0.000	192.168.1.70	172.217.110.4	QUIC	1276	Protected Payload (PP)
100	0.000	192.168.1.70	172.217.110.4	QUIC	78	Protected Payload (PP)
101	0.000	192.168.1.70	172.217.110.4	QUIC	66	Protected Payload (PP)

Frame 80: Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface hpf_10000000-0000-0000-0000-000000000000, 14 e

Ethernet II, Src: Intel_E7160-00 (08:00:27:00:00:00), Dst: Realtek_8100 (08:00:27:00:00:00)

Internet Protocol Version 4, Src: 192.168.1.70, Dst: 172.217.110.4

User Datagram Protocol, Src Port: 54581, Dst Port: 443

Source Port: 54581

Destination Port: 443

Length: 89

Checksum: 0x0000 (unverified)

Checksum Status: Unverified

[Stream Index: 40]

[Stream Packet Number: 20]

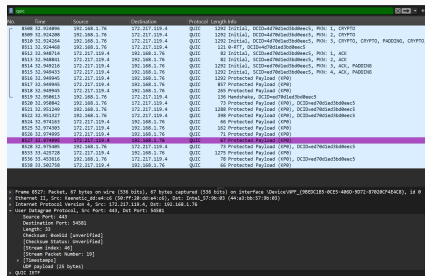
[Timestamps]

PP Payload (73 bytes)

QUIC HTTP

Рис. 21: Исходящий QUIC

Порты в обратном порядке: 443 → 54581.



No.	Time	Source	Destination	Protocol	Length	Info
8088	32.824886	292.168.1.76	172.217.119.4	QUIC	1292	Initial, DCID=67b0c1ed3b09ec5, Pkts: 1, CRYPTO
8089	32.824888	292.168.1.76	172.217.119.4	QUIC	1292	Initial, DCID=67b0c1ed3b09ec5, Pkts: 2, CRYPTO
8518	32.924264	292.168.1.76	172.217.119.4	QUIC	1292	Initial, DCID=67b0c1ed3b09ec5, Pkts: 3, CRYPTO, CRYPTO, PADDING, CRYPTO...
8511	32.924468	292.168.1.76	172.217.119.4	QUIC	121	ACK, DCID=67b0c1ed3b09ec5
8513	32.948714	172.217.119.4	292.168.1.76	QUIC	82	Initial, DCID=67b0c1ed3b09ec5, Pkts: 1, ACK
8519	32.948841	172.217.119.4	292.168.1.76	QUIC	82	Initial, DCID=67b0c1ed3b09ec5, Pkts: 2, ACK
8514	32.948216	172.217.119.4	292.168.1.76	QUIC	1292	Initial, DCID=67b0c1ed3b09ec5, Pkts: 3, ACK, PADDING
8515	32.948435	172.217.119.4	292.168.1.76	QUIC	1292	Initial, DCID=67b0c1ed3b09ec5, Pkts: 4, ACK, PADDING
8516	32.948940	172.217.119.4	292.168.1.76	QUIC	1292	Protected Payload (CNO)
8517	32.948940	172.217.119.4	292.168.1.76	QUIC	837	Protected Payload (CNO)
8518	32.948940	172.217.119.4	292.168.1.76	QUIC	589	Protected Payload (CNO)
8519	32.958613	292.168.1.76	172.217.119.4	QUIC	136	Handshake, DCID=67b0c1ed3b09ec5
8520	32.958842	292.168.1.76	172.217.119.4	QUIC	75	Protected Payload (CNO), DCID=67b0c1ed3b09ec5
8521	32.959146	292.168.1.76	172.217.119.4	QUIC	1288	Protected Payload (CNO), DCID=67b0c1ed3b09ec5
8522	32.959127	292.168.1.76	172.217.119.4	QUIC	888	Protected Payload (CNO), DCID=67b0c1ed3b09ec5
8524	32.974545	172.217.119.4	292.168.1.76	QUIC	86	Protected Payload (CNO)
8525	32.974593	172.217.119.4	292.168.1.76	QUIC	162	Protected Payload (CNO)
8526	32.974995	172.217.119.4	292.168.1.76	QUIC	78	Protected Payload (CNO)
8447	32.974995	172.217.119.4	292.168.1.76	QUIC	87	Protected Payload (CNO)
8528	32.975488	292.168.1.76	172.217.119.4	QUIC	78	Protected Payload (CNO), DCID=67b0c1ed3b09ec5
8529	32.421720	172.217.119.4	292.168.1.76	QUIC	1279	Protected Payload (CNO)
8536	32.453616	292.168.1.76	172.217.119.4	QUIC	76	Protected Payload (CNO), DCID=67b0c1ed3b09ec5
8538	32.580758	172.217.119.4	292.168.1.76	QUIC	66	Protected Payload (CNO)

Frame 8227: Packet, 87 bytes on wire (536 bits), 87 bytes captured (536 bits) on interface \Device\NPF{...} (0.0.0.0), id 0
Ethernet II, Src: Realtek-8000-00-00-00-00-00, Dst: Intel-82-57-00-03 (44:83:00:57:00:03)
Internet Protocol Version 4, Src: 172.217.119.4, Dst: 292.168.1.76
User Datagram Protocol, Src Port: 443, Dst Port: 54581
Stream: 0
Destination Port: 54581
Length: 87
Checksum: 0x0000 [unverified]
[Checksum status: unverified]
[Stream Index: 0]
[Stream Packet Number: 19]
[Timestamp]
UDP payload (87 bytes)
QUIC HTTP

Рис. 22: Входящий QUIC

Клиент 192.168.1.76:57608 → 13.249.8.110:443, флаг SYN, seq=0.

No.	Time	Source	Destination	Protocol	Length	Info
207	0.04513	13.249.8.139	13.249.8.139	TCP	232	Server-Mails, Charge Cipher, Encrypted Handshake Message
208	0.04521	13.249.8.139	13.249.8.139	TCP	44	Charge Cipher Index, Encrypted Handshake Message, Application Data
209	0.04526	13.249.8.139	5.255.255.242	TCP	164	Application Data
210	0.17043	13.249.8.139	5.255.255.242	TCP	88	Application Data
211	0.17526	5.255.255.242	13.249.8.139	TCP	4545	5800 [ACK] Seq=781 Win=893 Len=66
212	0.17685	5.255.255.242	13.249.8.139	TCP	585	Application Data
213	0.18152	5.255.255.242	13.249.8.139	TCP	44	Application Data
214	0.18157	13.249.8.139	5.255.255.242	TCP	88	50383 > 403 [ACK] Seq=140444444 Win=65535 Len=0
215	0.18542	13.249.8.139	13.249.8.139	TCP	66	403 > 57067 [ACK] Seq=140444527 Win=6906 Len=0
216	0.18547	13.249.8.139	13.249.8.139	TCP	422	Application Data
217	0.18689	13.249.8.139	13.249.8.139	TCP	66	57067 > 403 [FIN, ACK] Seq=527 Ack=140444503 Len=0
218	0.18914	13.249.8.139	13.249.8.139	TCP	76	5800 > 57067 [ACK] Seq=140444503 Len=0
219	0.18917	13.249.8.139	13.249.8.139	TCP	44	403 > 57067 [FIN, ACK] Seq=140444503 Len=0
220	0.19159	13.249.8.139	13.249.8.139	TCP	66	57067 > 403 [ACK] Seq=140445012 Win=6904 Len=0
221	0.19162	13.249.8.139	13.249.8.139	TCP	66	57067 > 403 [ACK] Seq=140445012 Win=6904 Len=0
222	0.35041	13.249.8.139	13.249.8.139	TCP	66	5800 > 403 [ACK] Seq=140445012 Win=6904 Len=0
223	0.35052	13.249.8.139	13.249.8.139	TCP	66	403 > 57067 (Seq=140445012)
224	0.35053	13.249.8.139	13.249.8.139	TCP	66	403 > 57067 [ACK] Seq=140445012 Win=6904 Len=0
225	0.35058	13.249.8.139	13.249.8.139	TCP	232	Server-Mails, Charge Cipher, Encrypted Handshake Message, Application Data
226	0.35060	13.249.8.139	13.249.8.139	TCP	232	Charge Cipher Index, Encrypted Handshake Message, Application Data
227	0.35058	13.249.8.139	13.249.8.139	TCP	66	403 > 57067 [ACK] Seq=140445000 Win=6906 Len=0
228	0.35073	13.249.8.139	13.249.8.139	TCP	64	Application Data
229	0.35073	13.249.8.139	13.249.8.139	TCP	66	57068 > 403 [ACK] Seq=140445000 Win=6906 Len=0
230	0.35079	13.249.8.139	13.249.8.139	TCP	66	403 > 57068 > 403 [ACK] Seq=140445000 Win=6906 Len=0
231	0.35099	13.249.8.139	5.255.255.242	TCP	76	403 > 57068 > 403 [ACK] Seq=140445000 Win=6906 Len=0
232	0.35102	13.249.8.139	5.255.255.242	TCP	47	Application Data
[Header checksum status: Verified]						
Source Address: 132.168.1.76						
Destination Address: 13.249.8.139						
[Stream index: 7]						
[Transmission Control Protocol, Src Port: 57068, Dst Port: 443, Seq: 0, Len: 0]						
Source Port: 57068						
Destination Port: 443						
Stream index: 131						
[Sequence Number: 5]						
[Conversation completeness: Incomplete, Data (55)]						
TCP segment len: 0						
Sequence Number: (relative sequence number)						
Sequence Number (raw): 173195450						
[Window Number: (relative sequence number)]						
Acknowledgment Number: 0						
Acknowledgment Number (raw): 0						
[Options (0 bytes), Window scale, No-Operation (NOP), Window scale, SACK permitted, Timestamps]						
[Flags: none, SYN]						

Рис. 23: SYN

SYN + ACK

Ответ сервера: SYN, ACK, seq=0, ack=1.

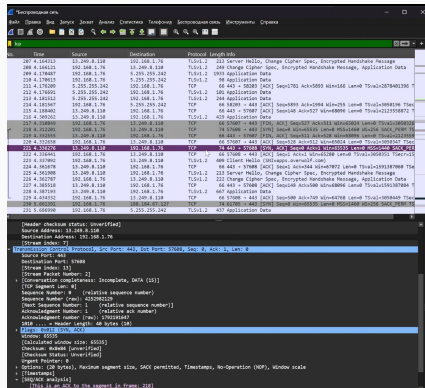


Рис. 24: SYN + ACK

Клиент подтверждает: только ACK, seq=1, ack=1.

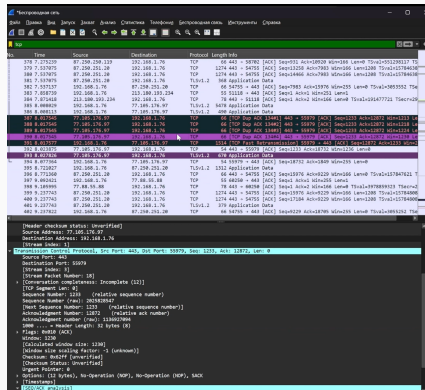


Рис. 25: АСК

Хендшейк на графике потока

Хорошо видны SYN, SYN+ACK, ACK перед GET.

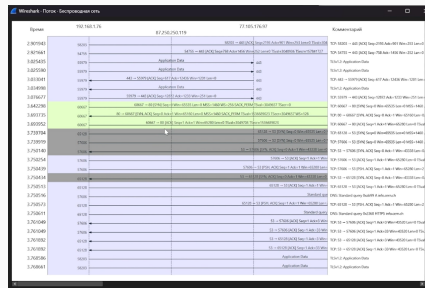


Рис. 26: Хендшейк на графике

Получены навыки анализа пакетов в Wireshark: разобраны кадры Ethernet, PDU транспортного и прикладного уровней стека TCP/IP.

1. Wireshark Foundation. Wireshark User's Guide [Электронный ресурс]. — URL: https://www.wireshark.org/docs/wsug_html_chunked/ (дата обращения: 07.10.2026).
2. Wireshark Foundation. Wireshark Display Filter Reference [Электронный ресурс]. — URL: <https://www.wireshark.org/docs/dfref/> (дата обращения: 07.10.2026).
3. IETF. RFC 826: An Ethernet Address Resolution Protocol [Электронный ресурс]. — URL: <https://datatracker.ietf.org/doc/html/rfc826> (дата обращения: 07.10.2026).